Clara: Semi-Automatic Annotation Algorithm for Medical Image Recognition

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***Abstract*—In the era of rapid development of artificial intelligence. With the help of artificial intelligence, various fields are developing rapidly. This paper discusses the potential of the combination of artificial intelligence and medical image to realize accurate medical image annotation and analysis. The accuracy of medical diagnosis depends on the accuracy of image recognition. In order to optimize image recognition, this paper proposes a semi-automatic image annotation method, which improves the professionalism of data through human-computer interaction and expands data rapidly. Our contribution is to improve the edge aliasing and imprecision of Live-Wire dimensions. It reduces the use of medical resources in the labeling process, improves the accuracy of data labeling, and ensures the accuracy of medical image recognition. This is a portable algorithm, and it can also improve the accuracy of image recognition in other fields. Image recognition experiments are carried out by using LabelMe and Live-Wire methods. The results show that Live-Wire method has a better recognition effect than LabelMe method, and the stability of the tag is more than 99%. To sum up, the combination of semi-automatic marking and medical image recognition provides a new possibility for medical application, and is expected to play an important role in the early diagnosis and treatment of diseases.**

***Index Terms*—image analysis, medical image processing, machine learning, data mining**

I. Introduction

This paper reviews the revolutionary changes that medical imaging technology has brought to disease diagnosis and treatment in recent years. With the emergence of artificial intelligence, it has great potential to combine machine learning technology with medical image diagnosis to improve the efficiency and accuracy of image recognition. However, the success of medical image recognition depends on the accuracy of medical imaging tags. And there have been some good results at present.

For example, A ResNet-based approach for accurate radiographic diagnosis of knee osteoarthritis published by Yu Wang et al. [1], this paper studies the recognition and classification of medical images based on artificial intelligence neural network, and obtains good results, but the accuracy is maintained at 81.41%, there is still room for improvement. Similarly, such as Two-view attention-guided convolutional neural network for mammographic image classification published by Lilei Sun et al in 2022 [2], we study the combination of neural network and image processing, considering that multiple views complement each other, so as to extract more kinds of feature information. To facilitate further medical analysis.

The above literature default to the impact of feature extraction accuracy on subsequent image processing, for feature extraction research, in the medical field is also a major difficulty, for this, there are related studies. For example, in 2022, Yonghong Zhang et al [3] proposed a two-stage method to locate lumbar segments, which mentioned the use of feature fusion technology to deal with the overall blank of the image to accurately locate, laying the foundation for the follow-up steps, so that the final results are better than other methods. However, in order to solve the problem of feature deviation, Dawei Dai et al [4] designed a different bias in different tasks in 2022, so as to get specifically expected features.

For this reason, this paper proposes a semi-automatic image annotation method to improve the image recognition [5] rate, which can significantly improve the accuracy of image recognition. This method can reduce the number of medical resources, ensure the efficiency and accuracy of medical diagnosis, and rapidly expand the amount of data through human-computer interaction.

The above research proves the importance of feature accuracy for medical image processing. For this requirement, based on the Live-Wire algorithm, this paper continues to optimize the edge accuracy and edge sawtooth elimination, and improve the speed, which greatly improves the accuracy of image recognition. The main contributions of this thesis are as follows: The main results are as follows: (1) the proposed semi-

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automatic image annotation method can significantly reduce the time and manpower required for image annotation, thus improving the efficiency of image recognition. This method can also significantly reduce the medical resources needed for image annotation. (2) We propose a loss mask and a complete mask to filter out irrelevant image features and improve the accuracy of image recognition. (3) the algorithm is portable, and the method can also be applied to computer vision, robots and other fields to improve the accuracy of image recognition.

Experiments were conducted using LabelMe and Live-Wire methods, and the results demonstrated that the Live-Wire [6] method significantly improves the annotation accuracy compared to the LabelMe method. Additionally, the classical image recognition algorithms Mask R-CNN [7] and BlendMask [8] were used to demonstrate the effectiveness of the proposed method in improving image recognition.

Generally speaking, the contribution of this paper will have a significant impact on the field of medicine. The semi-automatic image annotation method proposed in this paper can greatly improve the accuracy and efficiency of image recognition and has a wide application prospect.

The paper first analyzes the background of image annotation and points out the application value of the semi-automatic charged annotation method in section I. The section II, introduces the related work of image annotation and classical image recognition algorithms. Section III discusses image annotation and recognition methods for medical images. Section IV compares and analyzes annotation and recognition methods. Finally, the paper summarizes its contributions in section V.

II. Related Work

The existing tags are based on time-consuming and laborious manual labeling tools, and many studies are to obtain tagged data from outsourcing organizations. However, due to the strong professional requirements of tagging data, some researchers have carried out small sample circular learning and basically realized automatic tagging. The accuracy of machine labeling is also determined by sample labeling. The number and accuracy of tags directly affect the accuracy of image processing. Therefore, improving the speed and accuracy of tags has become a research goal. Based on the above research status, and based on the analysis of the existing annotation methods, a semi-automatic annotation method is proposed in this paper.

A. Image Annotation Method

1) Traditional Annotation Methods and Tools

: SVMMN [9] is a novel image annotation algorithm suggested by Liu. In criterion functions, SVMMN combines example smoothness and class smoothness. It provides a large margin while simultaneously reducing the number of samples in the large margin zone. Fluid annotation, developed by Andriluka [10] et al. is a user-friendly human-machine collaboration interface for annotating class labels and Outlines of each item and background region in an image. Niu [11] proposes a large-scale image annotation that addresses two issues: 1) how to develop rich feature representations for predicting a variety of visual notions from objects and scene abstract concepts, and 2) how to annotate photographs with the optimum number of annotations.

Other annotation tools include LabelMe [12], lablelimg and so on, all of which have common shortcomings. First, the annotation tool needs many manual labeling of each image. Second, the labeling process is tedious and time-consuming work and requires enough patience from the labeling personnel. Finally, image labeling requires strong professionalism. If the personnel are not familiar with the professional field, the labeling results may have a large deviation, resulting in inaccurate labeling. This paper studies these shortcomings, and the labeling efficiency is improved by more than 60%, and the accuracy is improved by about 10%.

2) The Full-automatic Labeling Algorithm

: Image annotation is critical to computer vision tasks, but manual labeling can be time-consuming and laborious. To save on human resources and reduce the time and space complexity of image annotation, various research institutes and institutions have developed different algorithms for image annotation and applied them to various fields.

Berriel [13] suggested a deep learning-based crosswalk categorization system. To further assist the autonomous driving area, the system employs crowd-sourcing platforms to automatically obtain and label large-scale data sets and trains convolutional neural networks for crosswalk categorization to propose an automatic labeling algorithm based on BIM technology [14], which was tested on pipeline welds and proved its automatic labeling impact. Cereal [15], an approach based on active learning, is offered. Humans only need to manually mark a few automatically picked regions in the corpus of unlabeled photos in this technique, which improves the effectiveness of semantic image segmentation algorithms while reducing human annotation effort.

The full-automatic labeling algorithm is based on deep learning neural network, which requires manual labeling. Automatic labeling algorithms still cannot avoid manual labeling. For example, Berriel's work still requires manual labeling, with an accuracy of 96.3% for manual labeling of specific areas of the database. Moreover, the automatic annotation algorithm lacks man-machine interaction and cannot correct the errors in image annotation in time. In this regard, the method described in this paper can be considered as correction in real-time, which not only improves the labeling speed but also increases the accuracy.

B. Image Recognition Algorithm

In the field of image recognition, labeling data is crucial for training image features. This paper proposes a semi-automatic annotation method called Live-Wire to label images in standard datasets and obtain high-precision training datasets. The labeled data is then incorporated into the image recognition model for training, and the training model is optimized to improve the accuracy of image recognition. Additionally, we introduce two classical image recognition algorithms.

1) Mask R-CNN Arithmetic
: Mask R-CNN is a general algorithm for image recognition. The model can recognize the target object in the image and mask it. Mask R-CNN is an improved version of Fast R-CNN [16], including the object Mask Prediction Branch in Parallel based on Fast R-CNN. Compared with other recognition algorithms, Mask R-CNN has a higher generalization ability. Moreover, a case segmentation algorithm combines target detection with semantic segmentation.

Mask R-CNN has been well applied in many fields. Carvalho et al. applied Mask R-CNN in strength segmentation of remote sensing images, the first study combined with multichannel remote sensing images. R. Anantharaman et al. applied Mask R-CNN in the detection and segmentation of oral diseases [17]. Kang Zhao et al. used Mask R-CNN with regularization of building boundaries to extract building images from satellite images [18].

The Mask R-CNN framework comprises three stages: the backbone network, RoIAlig, and three branches. First, the ResNet101 and Feature Pyramid Network(FPN) are the backbone feature extraction networks. ResNet101 adopts a cross-layer connection to extract image features, which deepens the depth of the neural network without reducing the performance of extracting features, making the results more accurate. Second, RoIAlig fine-tunes the 4 coordinates of each window obtained by RPN through bilinear interpolation and obtains a fixed coordinate pixel value, which makes the discontinuity continuous and reduces the error when the result returns to the original image [19]. Third, the ROI outputs by RPN were mapped to the shared feature map to extract the corresponding target features, which are output to FC and fully convolutional network (FCN), for instance, segmentation and target classification.

2) BlendMask Arithmetic
: BlendMask is a dense instance segmentation method incorporating top-down and bottom-up approaches. It implements instance segmentation via an attention technique based on the Fully Convolutional One-Stage(FCOS) [20]. The BlendMask module consists of three parts: the bottom module is used to process the underlying features, and the generated score map is called Base; the top layer is concatenated on the box head of the detector to generate the corresponding Base top-level attention; the blender module fuses The bases and attention to obtain the final forecast. Chen H et al. coupled instance-level information with finer-grained semantic information to increase the prediction accuracy of BlendMask's work, including the detector and BlendMask modules.

Compared to the Mask RegionProposal CNN's Network(R-CNN-RPN), the Detector Module employs FCOS directly, which minimizes the computation. Because BlendMask employs higher resolution and better information fusion features than Mask R-CNN, the mask created by BlendMask has greater quality and a more consistent inference time. BlendMask may also be included in other detection algorithms in various ways. Currently, the Blendmask is also meaningfully integrated into many areas. Quan L et al. [21] proposed a weed segmentation method based on BlendMask and effectively applied the deep learning method to weed leaf age and plant center identification, with the highest accuracy of 0.957.

III. Method

Image recognition technology plays a vital role in improving the efficiency and accuracy of disease recognition and enabling it to locate diseases. The importance of accurate labels as the foundation of image recognition is self-evident.

However, current labeling tools rely heavily on time-consuming and laborious manual work, and many researchers obtain labeled data from outsourcing organizations. Although some researchers have implemented small sample circular learning to achieve automatic tagging, the accuracy of machine labeling still depends on the quality of the sample labeling. The number and accuracy of tags directly impact image processing accuracy, making improving the speed and accuracy of labeling a critical research goal.

Based on the current research status and the analysis of existing annotation methods, this paper proposes a semi-automatic annotation method, Live-Wire, to label standard datasets with high precision, which is then used to train image recognition models to improve accuracy.

A. Improvement Based on Live-Wire

Live-Wire is a semi-automatic image segmentation tool with strong universality, leading to some shortcomings in the medical field. For example, the cost function of the traditional Live-wire method covers the characteristics of the edge point, the gradient amplitude of the edge point, and the change of the gradient direction between the nodes.

In this paper, $f_T(p,q)$ represents the gradient amplitude change feature between nodes $pandq$, and q node is the pixel point in the neighborhood of node p. If the gradient amplitude change of nodes p and q is more drastic, $f_T(p,q)$ is smaller. The path cost is more minor, and vice versa. $f_T(p,q)$ proposed in this paper is shown in Formula (1) :

$$f_T(p,q) = 1 - 1/e^{|G_p - G_q|} \tag{1}$$

Where $G(p), G(q)$ are the gradient amplitudes of point p and point q respectively, and are shown in Formula (2)

$$G(p) = \sqrt{I_x(p)^2 + I_y(p)^2}, G(q) = \sqrt{I_x(q)^2 + I_y(q)^2} \tag{2}$$

Suppose the cost function after optimization is shown in Formula (3) :

$$L'(p,q) = w_Z f_Z(q) + w_G f_G(q) + w_D f_D(p,q) + w_T f_T(p,q) \tag{3}$$

Where w_Z, w_G, w_D, w_T root offering [22] is set to $0.35, , 0.35, 0.15, 0.15$. $f_Z(q), f_G(q), f_D(p,q)$ are shown the Same as the traditional cost function, respectively to describe the Laplacian cross zero value, pixel gradient value, and pixel gradient direction.

In order to further suppress the discontinuity of edge extraction and improve the smoothness of edge extraction, the substitution function is changed by adding the variable ladder characteristic. It makes medical image labeling more accurate and convenient for the accurate recognition of medical images.

B. Application of Live-Wire

To meet the demand for image annotation various traditional annotation tools are adopted by major research institutions to meet the demand for different image annotations. LabelMe is an image annotation tool created by MIT's COMPUTER Science and Artificial Intelligence Laboratory (CSAIL) in 2010. VGG Image Annotator-VIA [23] was presented by Dutta et al. as a simple and independent manual image annotation tool in 2019. Quick Annotator [24] (QA), an open-source tool introduced by Miao's team in 2021, seeks to enhance the annotation efficiency of organizational structures by order of magnitude.

Despite the advantages of LabelMe, it still requires many manual marking points to mark every point for target objects with irregular edges, significantly increasing the complexity of the labeling process.

The Live-Wire method is an interactive, contour-based segmentation technology. In general, the contour trajectory of the object is constructed by selecting control points and finding the shortest path between them. The implementation of the algorithm only needs to select a seed point and a target endpoint at the edge of the target object, and then a path with the minimum cost from the seed point to the target endpoint can be searched according to the cost function and the Dijkstra algorithm.

Live-Wire has achieved great success in image segmentation. For example, Peyman Gholami [25] in 2017, We designed a new method of chronic wound healing based on semi-automatic wound image segmentation using Live-Wire, which increases the clinician's control over the print process by providing more precise coordinates. In 2018, David group [26] provided a semi-automatic technique for incorporating the right ventricular area and shape information into the Live-Wire framework. A slice segmentation result was utilized to segment neighboring slices. Yang Liu [9] wanted to assess students' soft and hard abilities in essential e-learning by using creative learning based on student-centered learning and problem-based learning models in 2018. Based on the excellent performance of Live-Wire in image segmentation, this paper considers applying Live-Wire to image annotation and recognition of medical images.

C. Experiment Comparison Calculation Method Description

1) Labeling Comparison Calculation Method

: In this paper, we label multiple target objects in different images using two different methods: traditional manual labeling with the LabelMe tool (method *I*), and the semi-automatic labeling method with the Live-Wire algorithm (method *II*). To quantitatively evaluate the stability and accuracy of the labeled samples, we use both methods to label targets in different images and calculate the ratio of stable label area pixels to the average value of total label area pixels in the same image. We selected five types of target objects for a comparative experiment, and the corresponding five images are shown in Fig.1 A-E.

The stable area ratio of the labeling algorithm can be calculated according to Formula (4)

$$r_{(k,m)} = \frac{n * S_{both(k,m)}}{\sum_{i=1}^{n} S_{(k,m,i)}} \tag{4}$$

Where $r_{(k,m)}$ is the proportion of stable regions in the multiple labeling of the k-th image (k = 1, 2, 3, 4, 5 figures A, B, C, D, E) labeled by the m-th (m = 1, 2 are Method I and Method II) methods. $S_{(k,m,i)}$ is the number of pixels occupied by the i-th sample after labeling when the m-th method is used to label the k-th image, and $S_{(bothk,m)}$ is the number of public pixels occupied by n labeled samples when the m-th method is used to label the k-th image.

2) Recognition Results Calculation Method

: In this paper, the annotation methods based on method *I* and method *II* are compared through the current classical image recognition algorithms to generate a high-quality mask and effective recognition in the case of good annotation speed and accuracy. Here, when calculating the recognition result, we mainly consider the COCO mask average precision (AP), AP at IoU 0.5 (AP_{50}), 0.75 (AP_{75}), and AP for objects at different sizes AP_S, AP_M and AP_L.

D. Experiments

This experiment is based on the Python 3.8 programming environment, detectron2, and PyTorch1.4 programming to achieve the above image recognition algorithm and test. Windows 10, GPU, cuda10 and cuDNN7.4 are used in the experiment. As shown in Table 1:

TABLE I
EXPERIMENTAL ENVIRONMENT

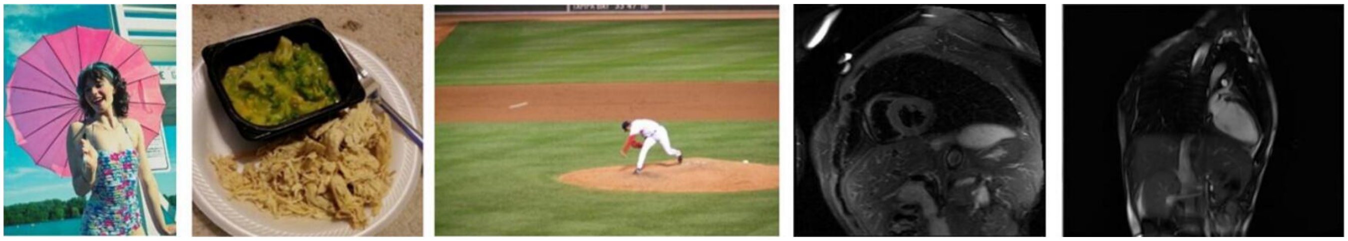
Operating System	Detectron Version	CUDA Version
Windows10	2	10
CUDNN Version	Python Version	Pytorch Version
7.4	3.8	1.4

The label annotation algorithm is realized by C++ programming and the image recognition algorithm is realized by Python programming. We use the original image of the coco data set to construct the data set and annotate the data set with method I and method II respectively, and then conduct image recognition processing.

E. Comparative analysis of the two annotation results

Labeling is the basic data processing of image recognition. However, aiming at the shortcomings of the traditional annotation tool Labelme, a semi-automatic annotation tool is proposed in this paper, and the experimental results are compared and analyzed.

Based on the above calculation method, we use method I and method II to label and experiment with multiple target objects in multiple images. The experimental results are analyzed qualitatively and quantitatively.



A Multi-node B Circular edge C Dynamic D Medical 1 E Medical 2

Fig. 1. The selected images for experiment

In view of the five pictures in Fig. 1, we use method I and method II to label them respectively and compare their labeling robustness.

The ratio calculation result is shown in Table 2:

TABLE II
ANALYSIS OF LABELING ROBUSTNESS OF METHOD I AND METHOD II SAMPLES

Picture	**MethodI** ratio	**MethodII** ratio
A	0.987085	0.997737
B	0.984244	0.993135
C	0.992371	0.997352
D	0.927436	0.972954
E	0.976543	0.994211

As shown in Table 2, the ratios of Method II are generally higher than the ratios of Method I. This means that the method using the LabelMe annotation tool has a lower degree of automation, its annotation is greatly affected by the artificially selected annotation points, its robustness is poor, and its accuracy is low. To improve the accuracy, we need to increase the manual marking points as much as possible, increase the labor cost and reduce the marking efficiency. The sample labeling method using the Live-Wire algorithm has higher robustness, better automatic labeling, higher efficiency and higher accuracy. The labeling result is shown in Fig.2-6:

For picture A, the result is shown in Fig.2. The edges of the umbrella are approximately polygonal. If we use Method I, we need to mark many marks in every endpoint of the polygons. But when we use Method II, we need only select a few seed points to finish labeling.

For picture B in Fig. 1, the results are shown in Fig. 3. The targets are more elliptical. In this case, if we use method I for labeling, we can only mark points as possible as close as possible and mark a mass of points. But if we use Method II, we only select a few seeds that are far apart, and then we can obtain the desired result. And method II observably enhances the degree of accuracy.

For Picture C, the result is shown in Fig. 4. The target is much smaller than the whole picture. Therefore if we use method I, we have to mark points more carefully and avoid mistakes. But when we have a large number of pictures to handle, we can't sure that we can finish the job as it says above. If we use Method II, we can just care about selecting some seed points and we can get a more reliable result.

From the perspective of medical data with high requirements for labeled images. the targets generally need to be marked to have clear borders and rich detail, and the percentage is quite modest. As seen in Fig. 5 and Fig. 6, We find it challenging to use Method I to quickly annotate anything effectively. Even though they only take up a little amount of space, we must mark as many points as we can in order to increase annotation accuracy. Method II is more suited for such intricate targets, though. Due to the boundary's complexity and smoothness, the method can quickly capture the complicated edge without the need for additional marking points for fitting. The whole border can only be marked by a small number of seed points, which increases production effectiveness and prevents a significant loss of material and human resources.

F. Analysis of image recognition results

At present, there are many excellent image recognition algorithms, such as Mask R-CNN and Mask R-CNN* (which is the modified Mask R-CNN with implementation details in TensorMask, TensorMask, and BlendMask. The experimental results of comparative experiments using these three methods are shown in Table 3.

The experimental data in Table 4 are from our preliminary image recognition results and the parameters of evaluation metrics in this table are COCO mask average precision (AP), AP at IoU 0.5 (AP_{50}), 0.75 (AP_{75}) and AP for objects at different sizes AP_S, AP_M and AP_L. It can be seen from the table that, compared with other algorithms, BlendMask and Mask R-CNN are the two fastest-running algorithms. BlendMask is the most efficient and uses fewer training iterations to obtain the highest average accuracy. The average accuracy of Mask R-CNN is not the best, but it has good performance and fast uptime. Therefore, we choose BlendMask for image recognition and use Mask R-CNN for auxiliary verification.

G. Analysis of image recognition results based on semi-automatic annotation

We apply Live-Wire to semi-automatic annotation. Experiments show that using the Live-Wire algorithm for image annotation can not only improve time efficiency but also obtain more stable and accurate label data. Obtained label data set is

A Original image B Label by LabelMe C Label by algorithm Live-Wire D Masking

Fig. 2. Annotation comparison of picture A in Fig.1

A Original image B Label by LabelMe C Label by algorithm Live-Wire D Masking

Fig. 3. Annotation comparison of picture B in Fig.1

A Original image B Label by LabelMe C Label by algorithm Live-Wire D Masking

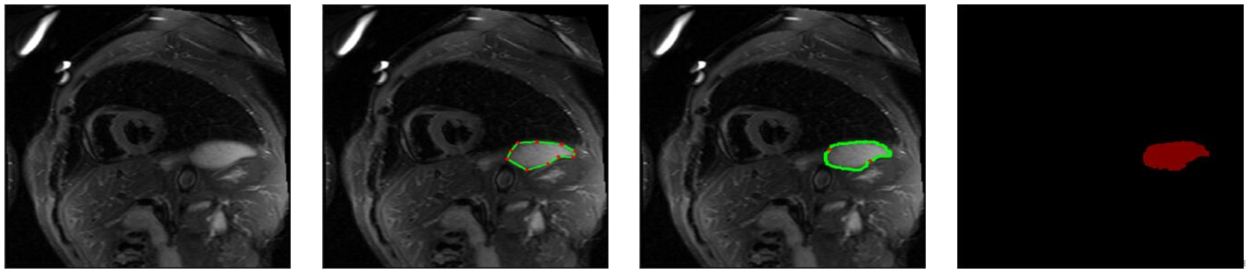
Fig. 4. Annotation comparison of picture C in Fig.1

TABLE III
DATA COMPARISON OF SEVERAL LATEST IMAGE RECOGNITION ALGORITHMS

Method	Backbone	Epochs	Time(ms)	AP	AP_{50}	AP_{75}	AP_S	AP_M	AP_L
TensorMask	5*R-50	72	400+	35.5	57.3	37.4	16.6	37.0	49.1
Mask R-CNN*		72	150+	36.8	59.2	39.3	17.1	38.7	52.1
Mask R-CNN		12	97.0	35.6	57.5	37.6	16.4	36.3	49.8
BlendMask		12	78.5	34.3	55.4	36.6	14.9	36.4	48.9
BlendMask		36	78.5	37.8	58.9	39.9	17.8	38.9	53.4

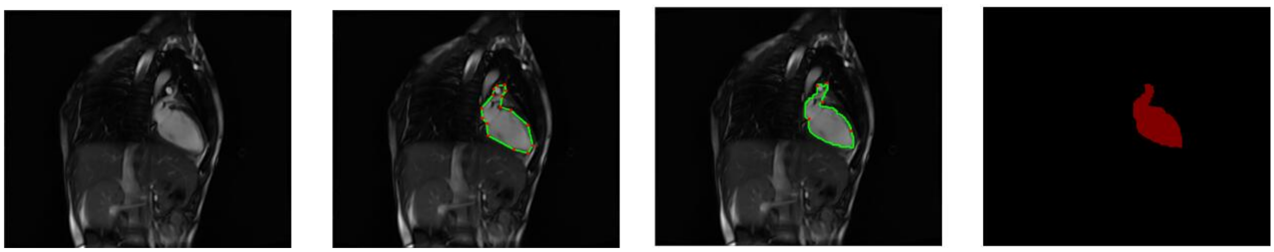
transmitted to the following learning model training, and the loss value and time of the model training are optimized, so that the expected better application effect can be achieved.

As shown in Fig. 7, figures a_1 and b_1 are the results of human face recognition through the BlendMask algorithm combined with method I annotation, and figures a_2 and b_2 are the results of human face recognition through the BlendMask algorithm combined with method II annotation. a_3 and b_3 pictures are close details. As shown in detail in Figure a_3, compared with the recognition result of the chin in Figure



A Original image | B Label by LabelMe | C Label by algorithm Live-Wire | D Masking

Fig. 5. Annotation comparison of the image D in Fig. 1



A Original image | B Label by LabelMe | C Label by algorithm Live-Wire | D Masking

Fig. 6. Annotation comparison of the image E in Fig. 1

11.png

a_1 **a_2** **a_3** **b_1** **b_2** **b_3**

Fig. 7. a_1 and b_1:BlendMask algorithm combined with method I. a_2 and b_2: BlendMask algorithm combined with method II. a_3 and b_3: Detail contrast.

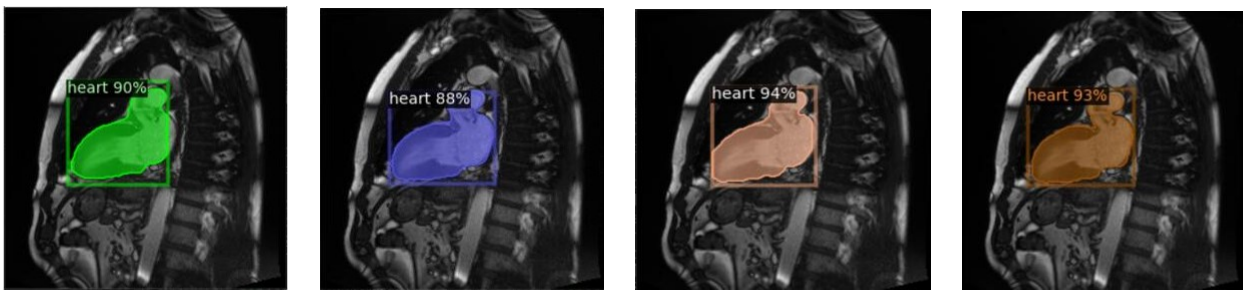
a_2, the recognition of the chin in Figure a_1 is missing. In contrast to a_2, the recognition of a_1's right forehead needs to be completed. As shown in detail in figure b_3, the right side of the person's face in figure b_1 has prominent incomplete recognition and a large area of blank space, while the person's face in figure b_2 is better recognized.

It is further shown that combining the BlendMask with method *II* is better than the combination with method *I* in dealing with edges. Through the comparison of the details, it is observed that the image edge of the data set labeled by method *II* is smoother, the recognition is more complete, and the accuracy is higher. In contrast, the data labeled by method *I* even have obvious recognition defects.

Finally, we compare the two annotation methods with two image recognition algorithms. As shown in Fig. 8 and Fig. 9. In Fig. 8, the first column shows the experimental results of the data set labeled by method I trained on Mask R-CNN; the second column is the experimental results of the data set labeled by method II trained on Mask R-CNN; the third column is the experimental results of data sets labeled by method I trained on BlendMask, and the fourth column is the experimental result of training on BlendMask

method I+ Mask R-CNN | method II+ Mask R-CNN | method I+ BlendMask | method II+ BlendMask

Fig. 8. Visualization of experimental results on general images



method I+ Mask R-CNN | method II+ Mask R-CNN | method I+ BlendMask | method II+ BlendMask

Fig. 9. Visualization of experimental results on medical images

using the data set labeled by method II. According to the experimental results, the prediction accuracy has reached the correct classification standard, so for the experimental results, we mainly consider the integrity of the recognition region and the accuracy of edge fit.

As shown in Fig.8, the combination of Mask R-CNN and method II can produce higher-quality masks. Compared with the Mask-RCNN algorithm, the recognition accuracy of the two methods is improved, and the recognition results of the data set annotated by method II are still smoother and more appropriate on the image edge, and the accuracy is also better.

Another example is the application of medical images. As shown in the last line of Fig. 9, similar situations also arise in the field of medical image segmentation, which necessitates high precision. There are nevertheless glaring discrepancies in the integrity, reducibility, and fit degree of the region edge, despite the fact that the prediction accuracy brought by various labeling methods is not different. The edges of the identification areas in method II + Mask R-CNN and method II + BlendMask have a higher degree of fitting and better integrity, which are more suitable for disciplines demanding high accuracy like medicine. In contrast, the left end of method I + Mask R-CNN and the lower left corner of figure method I + BlendMask exhibit apparent deviations.

It is further shown that the combination of the BlendMask with method II is better than the combination with method I in dealing with edges. Through the comparison of the details, it is observed that the image edge of the data set labeled by method II is smoother, the recognition is more complete and the accuracy is higher, while the data labeled by method I even have obvious recognition defects.

This paper mainly compares the quality of the mask generated by the two algorithms. Comparing the second column and the fourth column, the quality of the mask predicted by BlendMask is much higher than that of Mask R-CNN, especially on the boundary of the jawline, and BlendMask's prediction is more accurate. Comparing the third column and the fourth column, the mask quality predicted by the BlendMask model trained by the labeled data set of method II is better than the mask quality predicted by the BlendMask model trained by the labeled data set of method I. The results show that the image recognition model is trained by the data set based on method II annotation, the predicted mask has higher quality at the edge of the instance, and the predicted results are more accurate.

H. Quantitative comparison of recognition effect

Method I and method II respectively label the same data set and get different tagging results, and then use Mask R-CNN algorithm and BlendMASK algorithm to train the two tagging results respectively. Taking the image recognition accuracy and loss value trained by the image recognition algorithm as reference indicators, the following experimental results are obtained, as shown in Table 4:

Obviously, the loss values of the combinations completed using method II are less than those of the combinations completed using method I and the performance is better.

The horizontal comparison analysis of the experimental results shows that for the same annotation method, the loss value of the Blend-Mask algorithm is better than that of the Mask-RCNN, and the accuracy is higher. Longitudinal comparative analysis shows that whether the Mask-RCNN algorithm or the BlendMask algorithm is used for image recognition, the recognition results of annotation completed by method II are better in terms of loss value and accuracy. Comprehensive comparative analysis shows that in the whole process of image annotation and image recognition, the combined performance of using method II's semi-automatic annotation algorithm for image annotation and the BlendMask algorithm for image recognition is better, and has a small loss value.

TABLE IV
EXPERIMENTAL DATA OF MASK R-CNN GROUP AND BLENDMASK GROUP

	Parameter	**Method1**	**Method2**
3*Mask R-CNN	loss_mask	0.190	0.111
	total_loss	0.630	0.359
	accuracy	0.910	0.960
3*BlendMask	loss_mask	0.150	0.135
	total_loss	1.490	1.270
	accuracy	0.930	0.964

I. Summary

We are based on the medical field, and considering the research of key data of artificial intelligence in the medical field. In this paper, different image recognition algorithms are used to recognize the labeled data obtained by different methods. Through detailed experimental analysis, image comparison, and quantitative analysis, it is verified that the data labels obtained by method II have the advantages of smoother image edges, higher recognition accuracy, and lower loss values in image recognition. Compared with traditional methods, the semi-automatic label annotation method can better serve various image recognition algorithms. In addition, it has a better application effect and higher efficiency.

IV. CONCLUSION

According to the current research speed, image recognition will soon play a greater role in the field of medicine and health. As a combination of artificial intelligence and medical field, medical image recognition can assist doctors in the integrated process of observation, diagnosis and feedback. The realization of these functions is inseparable from some important means of artificial intelligence, especially the foundation of artificial intelligence, the pre-treatment of big data. In this paper, an improved semi-automatic labeling algorithm is proposed and compared with traditional labeling tools. Then, the semi-automatic labeling algorithm is applied to various image recognition algorithms, qualitative and quantitative analysis is carried out, and the visualization results are analyzed and explained in detail. Experimental results show that the image recognition method proposed in this paper has the best recognition effect from data processing to image recognition, especially for medical images, the recognition accuracy can reach more than 0.99. Although our method has achieved good results in terms of marking efficiency, accuracy and image recognition. However, in the huge amount of data, semi-automatic labeling can no longer meet the needs of scientific research. Therefore, in the future, we will consider the combination of artificial intelligence and nano-scale image recognition, as well as the marking of nano-medical images, which can save doctors' time and solve more difficult and complicated diseases.

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